ANSWERS OF MODEL TEST PAPER 3

INTERMEDIATE: GROUP – II

PAPER – 6: FINANCIAL MANAGEMENT & STRATEGIC MANAGEMENT

PAPER 6A : FINANCIAL MANAGEMENT

PART I – Case Scenario based MCQs

1. 1. (c) Calculation of cost of capital

Capital	Weight	Cost	Product
Debt	0.3	10%	3.00%
Preference	0.2	11%	2.20%
Equity	0.5	15%	7.50%
	Ko=		**12.70%**

2. (a)

3. (c)

4. (d)

5. (a)

Calculation of CFAT

Year		1	2	3	4	5	6
A) No. of quick deliveries p.d.		10,000	12,000	13,800	15,180	15,939	15,939
B) No. of overnight deliveries p.d.		2,000	2,400	2,760	3,036	3,188	3,188
C) No. of quick deliveries p.a.		36,50,000	43,80,000	50,37,000	55,40,700	58,17,735	58,17,735
D) No. of overnight deliveries p.a.		7,30,000	8,76,000	10,07,400	11,08,140	11,63,547	11,63,547
E) Chargeable quick deliveries		18,25,000	21,90,000	25,18,500	27,70,350	29,08,868	29,08,868
F) No. of delivery partners	1.5x(A+B)/30	600	720	828	911	956	956
Revenue (in crores)							
From quick deliveries (QD)	(E x 40)	7.30	8.76	10.07	11.08	11.64	11.64
From QD seller commission	(C x 700 x 5%)	12.775	15.330	17.630	19.392	20.362	20.362
From Overnight delivery subscription	(B/2 x 5000)	0.500	0.600	0.690	0.759	0.797	0.797

From OD seller commission	(C x 750 x 7%)	3.83	4.60	5.29	5.82	6.11	6.11
Total Revenue		**24.41**	**29.29**	**33.68**	**37.05**	**38.90**	**38.90**
Cost (in crores)							
Advertising		7	8	10	0	0	0
IT and customer care		8	8	8	8	8	8
Delivery partner salary	(F x 15000)	0.90	1.08	1.24	1.37	1.43	1.43
Delivery partner commission	(C+D) x 20	8.76	10.51	12.09	13.30	13.96	13.96
Depreciation	on investment in year 0	6	6	6	6	6	
	on investment in year 2		4	4	4	4	4
	on investment in year 4				5	5	5
Total Cost		**30.66**	**37.59**	**41.33**	**37.66**	**38.40**	**32.40**
PBT		-6.25	-8.30	-7.65	-0.61	0.51	6.51
Less: Tax		1.56	2.08	1.91	0.15	-0.13	-1.63
PAT		-4.69	-6.23	-5.74	-0.46	0.38	4.88
Add: Depreciation		6.00	10.00	10.00	15.00	15.00	9.00
CFAT		**1.31**	**3.77**	**4.26**	**14.54**	**15.38**	**13.88**

Computation of NPV

Year	Particulars		Cash Flows (in crores)	PVF @ 12.7%	PV (in crores)
0	Investment		-30	1	-30
1	Investment		-20	0.887	-17.75
3	Investment		-25	0.699	-17.46
1	Operating CFAT		1.31	0.887	1.16
2	Operating CFAT		3.77	0.787	2.97
3	Operating CFAT		4.26	0.699	2.98
4	Operating CFAT		14.54	0.620	9.01
5	Operating CFAT		15.38	0.550	8.46
6	Operating CFAT		13.88	0.488	6.77
6	Sale Proceeds	(30+20+25)x2	150	0.488	73.21
	NPV				**39.35**

2. **(d)** FL=%change in NP/%change in EBIT=6.9/6=1.15

3. **(c)** Since IRR of projects of company is greater than its cost of capital, the company should retain all ist earnings i.e. DPR = 0. As per walter Po = [0 + (0.15/0.125)10]/0.125 = 96

4. **(d)** 180 days

PART II – Descriptive Questions

1. **(a)** Determination of specific costs

(i) Cost Debt (K_d) $= \dfrac{\text{Interest}(1-t)+\dfrac{(RV-NP)}{N}}{\dfrac{(RV+NP)}{2}} = \dfrac{₹11(1-0.35)+\dfrac{(₹100-₹96)}{10\,\text{years}}}{\dfrac{(₹100+₹96)}{2}}$

$= \dfrac{₹7.15+₹0.4}{₹98}$ = 0.077 or 7.70%

(ii) Cost of Preference Shares (K_p) = $\dfrac{PD+\dfrac{(RV-NP)}{N}}{\dfrac{(RV+NP)}{2}} = \dfrac{₹12+\dfrac{(₹100-₹95)}{10\,\text{years}}}{\dfrac{(₹100+₹95)}{2}}$

$= \dfrac{₹12+₹0.5}{₹97.5}$ = 0.1282 or 12.82%

(iii) Cost of Equity shares (K_e) = $\dfrac{D_1}{P_0}+G = \dfrac{₹2}{₹22-₹2}+0.07$ = 0.17 or 17%

I – Interest, t – Tax, RV- Redeemable value, NP- Net proceeds, N- No. of years, PD- Preference dividend, D_1- Dividend at the end of the year, P_0- Price of share (net)

Using these specific costs we can calculate the book value and market value weights as follows:

(a) Weighted Average Cost of Capital (K_0) based on Book value weights

Source of capital	Book value (BV)	Specific cost (k) (%)	Total costs [BV (×) k]
Debentures	₹ 8,00,000	7.7	₹ 61,600
Preferences shares	2,00,000	12.8	25,600
Equity shares	10,00,000	17.0	1,70,000
	20,00,000		2,57,200
K_0 = ₹ 2,57,200/₹ 20,00,000 = 12.86 per cent			

(b) Weighted Average Cost of Capital (K_0) based on market value weights

Source of Capital	Market Value (MV)	Specific cost (k) (%)	Total costs [MV (×) k]
Debentures	₹ 8,80,000	7.7	₹ 67,760,
Preference shares	2,40,000	12.8	30,720
Equity shares	22,00,000	17.0	3,74,000
Total capital	33,20,000		4,72,480
K_0 = ₹ 4,72,480/₹ 33,20,000 = 14.23 per cent			

(b) Total Assets = ₹ 400 crores

Total Asset Turnover Ratio = 2.5

Hence, Total Sales = 400 × 2.5 = ₹ 1000 crores

Computation of Profits after Tax (PAT)

	(₹ in crores)
Sales	1000
Less: Variable operating cost @ 65%	650
Contribution	350
Less: Fixed cost (other than Interest)	80
EBIT	270
Less: Interest on debentures (15% × 200)	30
EBT	240
Less: Tax 40%	96
EAT	144

(i) Earnings per share

$$\therefore \text{ EPS} = \frac{₹\ 144 \text{ crores}}{10 \text{ crore equity shares}} = ₹\ 14.40$$

(ii) Operating Leverage

$$\text{Operating leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{350}{270} = 1.296$$

It indicates the choice of technology and fixed cost in cost structure. It is level specific. When firm operates beyond operating break-even level, then operating leverage is low. It indicates sensitivity of earnings before interest and tax (EBIT) to change in sales at a particular level.

(iii) Financial Leverage

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{270}{240} = 1.125$$

The financial leverage is very comfortable since the debt service obligation is small vis-à-vis EBIT.

(iv) Combined Leverage

Combined Leverage = $\frac{\text{Contribution}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{EBT}}$ Or Operating Leverage x Financial Leverage

= 1.296 × 1.125 = 1.458

The combined leverage studies the choice of fixed cost in cost structure and choice of debt in capital structure. It studies how sensitive the change in EPS is vis-à-vis change in sales.

(c) Working notes:

1. Computation of Current Assets and Current Liabilities:

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = \frac{2.5}{1} \text{ or } \frac{\text{Current Assets}}{2.5} = \frac{\text{Current Liabilities}}{1} = \text{k (say)}$$

Or, Current Assets = 2.5 k and Current Liabilities = k

Or, Working capital = (Current Assets – Current Liabilities)

Or, ₹2,40,000 = k (2.5 – 1) = 1.5 k

Or, k = ₹1,60,000

∴ Current liabilities = ₹1,60,000

Current assets = ₹1,60,000 × 2.5 = ₹4,00,000

2. Computation of Inventories

$$\text{Liquid ratio} = \frac{\text{Liquid assets}}{\text{Current liabilities}}$$

$$\text{Or, } 1.5 = \frac{\text{Current assets - Inventories}}{₹1,60,000}$$

Or, 1.5 × ₹1,60,000 = ₹4,00,000 – Inventories

Or, Inventories = ₹1,60,000

3. Computation of Proprietary fund; Fixed assets; Capital and Trade payables

$$\text{Proprietary ratio} = \frac{\text{Fixed assets}}{\text{Proprietary fund}} = 0.75$$

∴ Fixed assets = 0.75 Proprietary fund

and Net working capital = 0.25 Proprietary fund

Or, ₹2,40,000/0.25	=	Proprietary fund
Or, Proprietary fund	=	₹9,60,000
and Fixed assets	=	0.75 proprietary fund
	=	0.75 × ₹9,60,000
	=	₹7,20,000

Capital	=	Proprietary fund – Reserves & Surplus
	=	₹9,60,000 – ₹1,60,000
	=	₹8,00,000
Trade payables	=	(Current liabilities – Bank overdraft)
	=	(₹1,60,000 – ₹40,000)
	=	₹1,20,000

Construction of Balance sheet

(Refer to working notes 1 to 3)

Balance Sheet as at 31st March, 2023

Liabilities	(₹)	Assets	(₹)
Capital	8,00,000	Fixed assets	7,20,000
Reserves & Surplus	1,60,000	Inventories	1,60,000
Bank overdraft	40,000	Current assets (other than inventories)	2,40,000
Trade Payables	1,20,000		
	11,20,000		11,20,000

2. (a)

Ascertainment of probable price of shares of Akash limited		
Particulars	**Plan (i) (If ₹ 4,00,000 is raised as debt) (₹)**	**Plan (ii) If ₹ 4,00,000 is raised by issuing equity shares (₹)**
Earnings Before Interest (EBIT)	3,60,000	3,60,000
20% on (14,00,000 +4,00,000)		
Less: Interest on old debentures @ 10% on 4,00,000	40,000 3,20,000	40,000 3,20,000
Less: Interest on New debt @ 12% on ₹ 4,00,000	48,000	——
Earnings Before Tax (After interest)	2,72,000	3,20,000
Less Tax @ 50%	1,36,000	1,60,000
Earnings for equity shareholders (EAIT)	1,36,000	1,60,000
Number of Equity Shares	30,000	40,000
Earnings per Share (EPS)	₹ 4.53	₹ 4.00

Price/ Earning Ratio	8	10
Probable Price Per Share (PE ratio x EPS)	₹ 36.24	₹ 40

Working Notes

		₹
1.	Calculation of Present Rate of Earnings	
	Equity Share capital (30,000x 10)	3,00,000
	10% Debentures $\left(40{,}000\times\frac{100}{10}\right)$	4,00,000
	Reserves and Surplus	7,00,000
		14,00,000
	Earnings before interest and tax (EBIT) given	2,80,000
	Rate of Present Earnings = $\frac{2{,}80{,}000}{14{,}00{,}000}\times100$	20%
2.	Number of Equity Shares to be issued in Plan	$\frac{4{,}00{,}000}{40}=10{,}000$
	Thus, after the issue total number of shares	30,000+ 10,000 = 40,000
3.	Debt/Equity Ratio if ₹ 4,00,000 is raised as debt: $\frac{8{,}00{,}000}{18{,}00{,}000}\times100$ = 44.44%	

As the debt equity ratio is more than 40% the P/E ratio shall be 8 in plan (i)

(b) In this case the company has paid dividend of ₹2 per share during the last year. The growth rate (g) is 5%. Then, the current year dividend (D_1) with the expected growth rate of 5% will be ₹ 2.10.

The share price is = $P_o = \frac{D_1}{K_e - g}$

$$= \frac{₹\,2.10}{0.155-0.05}$$

$$= ₹\ 20$$

In case the growth rate rises to 8% then the dividend for the current year. (D_1) would be ₹ 2.16 and market price would be-

$$= \frac{₹\ 2.16}{0.155-0.08}$$

$$= ₹\ 28.80$$

In case growth rate falls to 3% then the dividend for the current year (D_1) would be ₹ 2.06 and market price would be-

$$= \frac{₹\ 2.06}{0.155 - 0.03}$$

= ₹16.48

So, the market price of the share is expected to vary in response to change in expected growth rate is dividends.

3. Statement showing Working Capital for each policy

(₹ in crores)

	Working Capital Policy		
	Conservative	Moderate	**Aggressive**
Current Assets: (i)	4.50	3.90	2.60
Fixed Assets: (ii)	2.60	2.60	2.60
Total Assets: (iii)	7.10	6.50	5.20
Current liabilities: (iv)	2.34	2.34	2.34
Net Worth: (v)=(iii)-(iv)	4.76	4.16	2.86
Total liabilities: (iv)+(v)	7.10	6.50	5.20
Estimated Sales: (vi)	12.30	11.50	10.00
EBIT: (vii)	1.23	1.15	1.00
(a) Net working capital position: (i)-(iv)	2.16	1.56	0.26
(b) Rate of return: (vii)/(iii)	17.3%	17.7%	19.2%
(c) Current ratio: (i)/(iv)	1.92	1.67	1.11

Statement Showing Effect of Alternative Financing Policy

(₹ in crores)

Financing Policy	**Conservative**	**Moderate**	**Aggressive**
Current Assets: (i)	3.90	3.90	3.90
Fixed Assets: (ii)	2.60	2.60	2.60
Total Assets: (iii)	6.50	6.50	6.50
Current Liabilities: (iv)	2.34	2.34	2.34
Short term Debt: (v)	0.54	1.00	1.50
Long term Debt: (vi)	1.12	0.66	0.16
Equity Capital	2.50	2.50	2.50
Total liabilities	6.50	6.50	6.50
Forecasted Sales	11.50	11.50	11.50
EBIT: (vii)	1.15	1.15	1.15
Less: Interest short-term debt : (viii)	0.06	0.12	0.18
	(12% of ₹ 0.54)	(12% of ₹ 1.00)	(12% of ₹ 1.50)
Long term debt : (ix)	0.18	0.11	0.03
	(16% of ₹ 1.12)	(16% of ₹ 0.66)	(16% of ₹ 0.16)

Earning before tax: (x)-(viii+ix)	0.91	0.92	0.94
Taxes @ 35%	0.32	0.32	0.33
Earning after tax: (xi)	0.59	0.60	0.61
(a) Net Working Capital Position: (i)-[(iv)+(v)]	1.02	0.56	0.06
(b) Rate of return on shareholders Equity capital:(xi)/Equity Capital	23.6%	24%	24.4%
(c) Current Ratio: [(i)/(iv)+(v)]	1.35%	1.17	1.02

4. **(a)** "The profit maximisation is not an operationally feasible criterion." This statement is true because Profit maximisation can be a short-term objective for any organisation and cannot be its sole objective. Profit maximization fails to serve as an operational criterion for maximizing the owner's economic welfare. It fails to provide an operationally feasible measure for ranking alternative courses of action in terms of their economic efficiency. It suffers from the following limitations:

(i) Vague term: The definition of the term profit is ambiguous. Does it mean short term or long term profit? Does it refer to profit before or after tax? Total profit or profit per share?

(ii) Timing of Return: The profit maximization objective does not make distinction between returns received in different time periods. It gives no consideration to the time value of money, and values benefits received today and benefits received after a period as the same.

(iii) It ignores the risk factor.

(iv) The term maximization is also vague.

(b) "Financing a business through borrowing is cheaper than using equity"

(i) Debt capital is cheaper than equity capital from the point of its cost and interest being deductible for income tax purpose, whereas no such deduction is allowed for dividends.

(ii) Issue of new equity dilutes existing control pattern while borrowing does not result in dilution of control.

(iii) In a period of rising prices, borrowing is advantageous. The fixed monetary outgo decreases in real terms as the price level increases.

(c) **Meaning of Weighted Average Cost of Capital (WACC):** The composite or overall cost of capital of a firm is the weighted average of the costs of the various sources of funds. Weights are taken to be in the proportion of each source of fund in the capital structure. While making financial decisions this overall or weighted cost is used. Each investment is financed from a pool of funds which represents the various sources from which funds have been raised. Any decision of investment, therefore, has to be made with reference to the overall cost of capital and not with reference to the cost of a specific source of fund used in the investment decision.

The weighted average cost of capital is calculated by:

(i) Calculating the cost of specific source of fund e.g. cost of debt, equity etc;

(ii) Multiplying the cost of each source by its proportion in capital structure; and

(iii) Adding the weighted component cost to get the firm's WACC represented by K_0.

$K_0 = K_1 W_1 + K_2 W_2 + \ldots\ldots\ldots$

Where,

K_1, K_2 are component costs and W_1, W_2 are weights.

OR

(c) Assumptions of Modigliani – Miller Theory

(a) Capital markets are perfect. All information is freely available and there is no transaction cost.

(b) All investors are rational.

(c) No existence of corporate taxes.

(d) Firms can be grouped into "equivalent risk classes" on the basis of their business risk.